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THE GROUND STATE OF MATTER AT HIGH DENSITIES: EQUATION OF STATE AND STELLAR MODELS*

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ABSTRACT

The equation of state of zero-temperature matter in complete nuclear equilibrium is given for mass densities below 5×10^{14} g cm^{-3}. We redetermine, taking into account the effect of the Coulomb lattice and using more recent nuclear mass extrapolations, the sequence of equilibrium nuclides present at mass densities between 10^4 and 4.3×10^{11} g cm^{-3}, the point of neutron drip; and calculate the equation of state here. For densities between 4.3×10^{11} and 5×10^{14} g cm^{-3} the equation of state is taken from the calculations of Baym, Bethe, and Pethick; included is an extension of the equation of state to still higher densities, taken from the calculations of Pandharipande, which include the effects of hyperons. We calculate zero-temperature white-dwarf and neutron-star models for the equation of state. The maximum stable white-dwarf mass, the Chandrasekhar limit, is 1.00 $M_\odot$. The lightest stable neutron star has a mass 0.0925 $M_\odot$ and a central density 1.55×10^{14} g cm^{-3}; neutron stars between 0.0925 and 0.11 $M_\odot$ are entirely solid. Moments of inertia, surface deformations, and mass quadrupole moments are calculated for slowly rotating neutron stars.

I. INTRODUCTION

The wealth of information on pulsars (see, e.g., Hewish 1970) that has been gathered so far provides considerable incentive for investigating in detail the nature of the matter composing neutron stars. For example, one would like to relate moments of inertia inferred from observations on the slowing down of the Crab pulsar and the energy emission of the nebula to particular stellar models; one would also like more fully to understand possible relations of the observed speedups of the Crab and Vela pulsars to their underlying neutron-star structure. To a first approximation the matter in neutron stars is in its ground state; its thermal energy is very small on the scale of typical excitation energies, and it may be assumed to be in nuclear equilibrium—that is, its energy cannot be lowered by changing its constituents via strong, weak, or electromagnetic interactions (Tsuruta and Cameron 1966).

The character of matter in its ground state changes considerably with increasing density (see, e.g., Ruderman 1969). At very low density, matter in its ground state consists of ^{56}Fe nuclei, arranged in a lattice so as to minimize their Coulomb interaction energy. As the matter density ρ increases, the electron chemical potential (or Fermi energy including rest mass), μ_e, increases. Above $\rho \sim 10^4$ g cm^{-3} the electrons are essentially free, and above $\sim 10^7$ g cm^{-3} the electrons are fully relativistic. For $\mu_e \gtrsim 1$ MeV, or $\rho \gtrsim 8 \times 10^6$ g cm^{-3}, ^{56}Fe matter is no longer the lowest energy state; the energy can be lowered by having the nuclei capture energetic electrons, lose energy via neutrino emission, and become more neutron-rich. In the region beyond the point where this first can happen, the lowest energy state of matter is a lattice of ^{62}Ni nuclei. With increasing mass density, the equilibrium nuclide present in the lattice becomes progressively richer in neutrons. Eventually, at a density $\rho \approx 4.3 \times 10^{11}$ g cm^{-3}, the "neutron drip" point, the nuclei are so neutron-rich that with further increase in density, the continuum neutron states begin to be populated. Between "neutron drip" and $\sim 2 \times 10^{14}$ g cm^{-3} the matter consists of nuclei in a lattice and, in addition, a gas of free

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neutrons as well as electrons penetrating the lattice. (We shall refer to matter in this range as being in the *free-neutron regime.*) Above $\sim 2 \times 10^{14}$ g cm^{-3}, nuclei are no longer present. Rather, the matter is a uniform mixture of primarily neutron, proton, and electron liquids; as the density is raised, muons and various strange particles such as Σ and Λ appear and coexist stably.

At densities below $\sim 10^4$ g cm^{-3} a fraction of the electrons are bound to the nuclei, which are ^{56}Fe. The major difficulty in deriving the equation of state is to calculate the electronic energy. The calculations of Feynman, Metropolis, and Teller (1949), who used the Thomas-Fermi method, provide a sufficiently accurate description of this regime for use in constructing zero-temperature stars. When the spacing between nuclei becomes small compared with the Thomas-Fermi radius r_{TF} of an isolated neutral atom, the electrons are effectively completely ionized. Since $r_{TF} \sim a_0 Z^{-1/3}$, where a_0 is the Bohr radius and Z the number of protons in the nucleus, one may assume complete ionization for $\rho \gg 6AZ$ g cm^{-3}, where A is the total number of nucleons in the nucleus; this density is about 10^4 g cm^{-3} for ^{56}Fe.

Between 10^4 and 4.3×10^{11} g cm^{-3}, where free neutrons begin to appear, the electrons are essentially free, and soon become relativistic. The equilibrium nuclide present becomes increasingly neutron-rich; the nuclei are stabilized against β-decay by the filled electron Fermi sea. To find the equation of state is a twofold job: one must both determine the nuclide present and calculate the equation of state for that nuclide. The main problem in this regime is the lack of knowledge of nuclear masses far from the valley of β-stability. Considerable work has been done on this problem in connection with the search for superheavy elements in the laboratory (see Nilsson *et al.* 1969; Marinov *et al.* 1971) and also in connection with the origin of the elements, but there remain great uncertainties in the extrapolations of the nuclear shell structure into the very neutron-rich regime. Recent extrapolations have been given by Myers and Swiatecki (1966), Garvey *et al.* (1969), and Seeger (1970). The principal contributions to the energy of matter in this regime are the nuclear energy, the free electron energy, and the (negative) lattice energy. Detailed calculations of the equation of state here have been given by Salpeter (1961). While the lattice contribution to the total pressure is a small effect, $\sim Z^{2/3} e^2/\hbar c$, it can play a significant role, at higher densities, in the determination of the equilibrium nucleus. The reason is that the equilibrium species of nucleus present is determined by a competition between the nuclear surface energy, which tends to favor nuclei with a large number of nucleons, A, and the Coulomb energy, which tends to favor small-A nuclei. The Coulomb energy is the sum of the positive nuclear Coulomb self-energy and the negative lattice Coulomb energy. At a density of 2×10^{11} g cm^{-3} the nuclei occupy $\sim 10^{-3}$ of space, and the effect of the lattice energy is to reduce the net Coulomb energy by ~ 15 percent; in the absence of nuclear shell effects this would increase the equilibrium nuclear size by ~ 15 percent. While Salpeter included the effect of the lattice on the total pressure, he neglected its effect in determining the equilibrium species of nucleus present. One purpose of this paper is to redetermine this sequence of nuclides, taking into account the effects of the lattice energy, and making use of more recent tables of extrapolated nuclear masses.

The equation of state of matter in the free-neutron regime has recently been discussed by Baym, Bethe, and Pethick (1971; hereafter referred to as BBP). The main problem in this regime, as before, is to find the masses of the nuclei. Not only are the nuclei very neutron-rich, but they are immersed in a sea of free neutrons which exert a pressure on the surface of the nuclei and also lower the nuclear surface energy. BBP described the nuclei by a compressible liquid-drop model designed to take into account these effects of the free neutrons. In this regime the lattice Coulomb energy becomes exceedingly important in determining the nuclear size, since the lattice Coulomb energy is almost equal in magnitude, and of opposite sign to the nuclear Coulomb self-energy. As the density increases, the nuclei fill more and more of space. Finally at a density $\sim 2.4 \times 10^{14}$ g cm^{-3} they begin to touch. With further increase in density the density inhomoge-

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neity begins to smooth out and then disappears discontinuously in a first-order phase transition at about 3×10^{14} g cm^{-3}. Below this density the uniform-liquid phase is unstable against density fluctuations. Muons begin to appear at about this point. The calculations of BBP did not take into account the effect of possible superfluidity on the equation of state (see Yang and Clark 1971), but this is expected to be small.

To discuss the equation of state in the uniform-liquid phase, one can employ the standard techniques of nuclear-matter theory with some confidence up to about 5×10^{14} g cm^{-3}. BBP give the equation of state, based on Siemens's (1971) neutron-matter calculations, up to this density. The equation of state beyond this density is very uncertain, since, on the one hand, one has little knowledge of the forces between the various hyperons present in the matter there and, on the other hand, the particle density is sufficiently high that the conventional techniques of nuclear matter theory begin to require modification. To acquire some feeling for the stellar models with central densities in this region, we shall explore the consequences of one extension given by Pandharipande (1971) of the equation of state to very high densities, namely, his model C. This calculation allows for the presence of n, p, e, μ^-, Λ, $\Sigma^\pm$, Σ^0, $\Delta^\pm$, and Δ^0 in the matter, and assumes that the interactions involving hyperons are given by the $T = 1$ component of the Reid soft core potential, modified by omitting the one-pion-exchange part and reducing by 10 percent the intermediate-range attraction. The interaction between pairs of nucleons is the Reid soft core potential. The equation of state was calculated by assuming a Jastrow-type wave function, including only up to two-body clusters.

In the next section of this paper we carry out the redetermination of the equilibrium nuclei and the equation of state between 10^4 g cm^{-3} and neutron drip. We turn then, in § III, to construct models of nonrotating, zero-temperature neutron stars and white dwarfs, by integrating the Tolman-Oppenheimer-Volkoff equation. Then, in § IV, we calculate various properties of slowly rotating stars, including the moment of inertia, the surface deformations, and the mass quadrupole moment.

II. THE EQUATION OF STATE

We first consider the problem of determining the nuclide present at any density before neutron drip. Let us assume that the matter is composed of nuclei with nucleon number A and charge Z; we let n_N be the number of nuclei per unit volume. The total energy of the system per unit volume before neutron drip is then

$$E_{\rm tot}(A, Z, n_N) = n_N(W_N + W_L) + E_e(n_e) , \tag{1}$$

where $W_N(A, Z)$ is the total energy of an isolated nucleus, including rest masses of the nucleons but not including any electron energy, and $E_e(n_e)$ is the total electron energy per unit volume, where

$$n_e = Zn_N \tag{2}$$

is the mean electron density. The term W_L is the lattice energy per nucleus; for a given density of nuclei of charge Z, the lattice energy is minimized for a body-centered cubic (bcc) lattice, and is given by (Coldwell-Horsfall and Maradudin 1960)

$$W_L = -1.819620Z^2e^2/a , \tag{3}$$

where the lattice constant a is related to n_N by

$$n_N a^3 = 2 . \tag{4}$$

Above 10^4 g cm^{-3} the electrons may be considered free, in which case

$$E_e = \int_0^{k_e} \frac{k^2 dk}{\pi^2} c(\hbar^2k^2 + m_e^2c^2)^{1/2}$$

$$= \frac{m_e^4c^5}{8\pi^2\hbar^3} \{(2t^2 + 1)t(t^2 + 1)^{1/2} - \ln [t + (t^2 + 1)^{1/2}]\} , \tag{5}$$

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where k_e is the electron Fermi wavenumber and

$$t = \hbar k_e / m_e c \,. \tag{6}$$

For completely relativistic electrons, $t \gg 1$, and

$$E_e = \tfrac{3}{4} n_e m_e c^2 t \,. \tag{7}$$

The equilibrium values of A and Z at any density are those that minimize $E_{\rm tot}$ at fixed density of baryons n_b. The densities of nuclei and electrons are given in terms of n_b by

$$n_N = n_b / A \,, \quad n_e = n_b Z / A \,. \tag{8}$$

We carried out this minimization by simply using a table of nuclear masses to find the nucleus A, Z that gives the minimum $E_{\rm tot}$ for fixed n_b. In Table 1, we show the nuclei that were candidates in the minimization. The nuclear energies used here were taken from the mass table of Myers and Swiatecki (1965); we have also repeated the calculation using the tables of Garvey *et al.* (1969), but this produced no significant change in the results. The equilibrium nuclides present are listed in Table 2.

The nuclear energies used here have the electron rest masses subtracted out, but not the atomic-electron binding energy. We have not attempted to subtract out this electron binding energy for the following reason. The electron density distribution in the solid is uniform to a first approximation; the interaction energy of this uniform distribution is included in the lattice energy W_L. However, because of electron-screening effects, there are slight deviations of the electron distribution from uniformity, which produce an energy (Dyson 1971)

$$W_{\rm screening} = -0.2535 Z^{7/3} \frac{m_e e^4}{\hbar^2} \frac{\mu_e}{m_e c^2} \tag{9}$$

per nucleus, where

$$\mu_e = \partial E_e / \partial n_e \tag{10}$$

is the electron Fermi energy including the rest mass. By comparison, the electron energy, not including rest masses, of an isolated atom is given, in the Thomas-Fermi approximation, by

$$W_{\rm atom} = -0.766 Z^{7/3} m_e e^4 / \hbar^2 \,, \tag{11}$$

of comparable size to $W_{\rm screening}$. By not subtracting out the electronic contribution to the binding energies we make an error of the same magnitude, or smaller, and of opposite sign as the neglect of $W_{\rm screening}$ in $E_{\rm tot}$. Furthermore, the contribution of these two terms to the equation of state is negligible.

The pressure P in a star increases continuously with increasing depth. This implies that an abrupt density discontinuity must accompany each change of nuclear species. We may estimate this discontinuity by noting that to a first approximation the pressure before neutron drip is that due to electron kinetic energy. Thus the electron number density is essentially continuous across each transition. However, the density of baryons is given by $n_e A/Z$, so that across a transition from nucleus (A, Z) to nucleus (A', Z') the baryon density changes by

$$n'_b - n_b \approx n_e \left(\frac{A'}{Z'} - \frac{A}{Z} \right) . \tag{12}$$

The fractional change in the mass density, $\rho = E_{\rm tot}/c^2$, is given approximately by

$$\frac{\Delta \rho}{\rho} \approx \frac{\Delta n_b}{n_b} \approx \frac{Z/A}{Z'/A'} - 1 \,. \tag{13}$$

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TABLE 1

NUCLEI TESTED IN ENERGY MINIMIZATION

Proton Number \ Neutron Number	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66	68	70	72	74	76	78	80	82	84	86	88	90
24													x	x	x																			
26			x	1^{1}	x	x	x	x	x				x	8	x																			
28					x	2^{2}	3^{3}	x^{4}	x^{5}	x	x		x	7	x																			
30							x	x	x	x	x	x^{6}	x^{7}	6^{8}	x	x	x																			
32										x	x	x	x	5	x	x	x	x^{9}	x	x	x	x	x							x				
34												x	x	4	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x				
36													x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	12				
38													x	x	x				x	x	x	x	x	x	x	x	x	x	x	11^{10}	x	x		
40																						x	x	x	x	x	x	x	x	10	x	x	x	
42																												x	x	9	x	x	x	x
44																													x	x	x	x	x	x
46																														x				
48																														x				

NOTE.—Filled-in spaces indicate nuclei tested. Nuclei that actually appear are indicated by integers that give the order in which the nuclei occur as the density increases. The superscript numbers indicate the sequence found by Salpeter (1961).

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TABLE 2

EQUILIBRIUM NUCLEI BELOW NEUTRON DRIP

Nucleus	b (MeV)	Z/A	ρ_{max} (g cm^{-3})	μ_e (MeV)	$\Delta\rho/\rho$ (%)
^{56}Fe.....	8.7905	0.4643	8.1×10^{6}	0.95	2.9
^{62}Ni.....	8.7947	0.4516	2.7×10^{8}	2.6	3.1
^{64}Ni.....	8.7777	0.4375	1.2×10^{9}	4.2	7.9
^{84}Se.....	8.6797	0.4048	8.2×10^{9}	7.7	3.5
^{82}Ge.....	8.5964	0.3902	2.2×10^{10}	10.6	3.8
^{80}Zn.....	8.4675	0.3750	4.8×10^{10}	13.6	4.1
^{78}Ni.....	8.2873	0.3590	1.6×10^{11}	20.0	4.6
^{76}Fe.....	7.9967	0.3421	1.8×10^{11}	20.2	2.2
^{124}Mo.....	7.8577	0.3387	1.9×10^{11}	20.5	3.1
^{122}Zr.....	7.6705	0.3279	2.7×10^{11}	22.9	3.3
^{120}Sr.....	7.4522	0.3166	3.7×10^{11}	25.2	3.5
^{118}Kr.....	7.2002	0.3051	(4.3×10^{11})	(26.2)	...

NOTE.—b is the binding energy per nucleon; ρ_{max} is the maximum density at which the nuclide is present; μ_e is the electron chemical potential at that density, and $\Delta\rho/\rho$ is the fractional increase in the mass density in the transition to the next nuclide. The value of $\rho_{max} = 4.3 \times 10^{11}$ g cm^{-3} is the density at which neutron drip begins.

For example, in the first transition from ^{56}Fe to ^{62}Ni the density increases by 2.9 percent.

Because of these density discontinuities, it is necessary to make a Maxwell equal-area construction in a plot of P versus $1/n_b$ to determine the pressure at which the transition takes place. A somewhat more convenient procedure is to treat the *pressure*, rather than the baryon density, as the independent variable; then, as Salpeter noted, the quantity to be minimized at fixed P is the baryon chemical potential

$$\mu \equiv \frac{E_{tot} + P}{n_b} . \tag{14}$$

The pressure is given by

$$P = n_b^2 \left.\frac{\partial(E_{tot}/n_b)}{\partial n_b}\right|_{Z,A} , \tag{15}$$

which becomes, on using equations (1) and (8),

$$P = P_e + \tfrac{1}{3} W_L n_N . \tag{16}$$

The first term is the electronic pressure, given by

$$P_e = n_e \frac{\partial E_e}{\partial n_e} - E_e , \tag{17}$$

while the second is the negative pressure of the Coulomb lattice. Thus

$$\mu = (W_N + \tfrac{4}{3} W_L + Z\mu_e)/A . \tag{18}$$

To perform the required minimization one assumes a pressure P, an A, and a Z, and solves (16) for the electronic density, using $n_N = n_e/Z$. Given n_e, μ is evaluated from equation (18) by using tabulated values of $W_N(A, Z)$. Though this method is more involved than the minimization of E_{tot} at fixed n_b, it is useful for investigating the transitions from one nuclide to another. Note that μ is continuous across a transition.

Table 2 gives the resulting sequence of nuclides and their binding energies, together with the maximum density and electron chemical potential at which they occur, and the

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fractional mass density increase across the transition to the next nuclide. The ratio Z/A decreases with increasing density. The values of the binding energy b per nucleon, defined by

$$W_N = m_n c^2(A - Z) + m_p c^2 Z - bA \,, \tag{19}$$

are taken from Myers and Swiatecki (1965). In the sequence of nuclides in Table 2, only the first three, ^{56}Fe, ^{62}Ni, and ^{64}Ni, are stable in the laboratory; the fourth, ^{84}Se, has a laboratory half-life of 3.3 minutes. The masses of the remaining nuclei are determined by extrapolating from laboratory nuclei. It is interesting to notice that the nuclei from ^{84}Se to ^{76}Fe have 50 neutrons, while beyond this the nuclei from ^{124}Mo to ^{118}Kr have 82 neutrons. These neutron numbers are both magic numbers for laboratory nuclei; that they turn out to be favorable here reflects the shell structure for neutron-rich nuclei assumed in the extrapolations. Whether or not 82 neutrons is a magic number for a nucleus as neutron-rich as ^{118}Kr remains an interesting theoretical problem. Preliminary nuclear model calculations made by Ravenhall and Bennett (1971), who used a Hartree-Fock approach with the δ-function pseudo-potential of Skyrme (1959) and the potential parameters determined by Vautherin and Brink (1970), answer this question in the affirmative.

The role of the lattice energy in determining the sequence of nuclides can be seen by comparing Table 2 with Table 3, in the calculation of which the lattice energy was left out in minimizing $E_{\rm tot}$ with respect to Z and A. While the nuclear shell structure tends to be dominant, the lattice energy does have the consistent effect of raising the value of A, as can be seen, for example, from the fact that the 82-neutron shell appears with the lattice at 1.9×10^{11} g cm^{-3}, and without it at 3.1×10^{11} g cm^{-3}. At a given density within a given neutron-shell sequence, the nucleus calculated by including the lattice has a higher A than the nucleus found by omitting the lattice. We also give Salpeter's original results in Table 4.

The fact that A and Z minimize

$$\frac{E_{\rm tot}}{n_b} = \frac{W_N + W_L}{A} + \frac{E_e(n_b Z/A)}{n_b} \tag{20}$$

TABLE 3

EQUILIBRIUM NUCLEI BELOW NEUTRON DRIP IF THE LATTICE ENERGY IS NEGLECTED

Nucleus	b (MeV)	Z/A	$\rho_{\rm max}$ (g cm^{-3})	μ_e	$\Delta\rho/\rho$ (%)
^{56}Fe	8.7905	0.4643	8.5×10^{6}	0.96	2.7
^{62}Ni	8.7947	0.4516	1.3×10^{8}	2.0	0.7
^{58}Fe	8.7922	0.4483	2.8×10^{8}	2.6	2.4
^{64}Ni	8.7777	0.4375	1.1×10^{9}	4.1	3.0
^{66}Ni	8.7404	0.4242	1.3×10^{9}	4.2	3.0
^{68}Ni	8.7038	0.4118	1.9×10^{9}	4.7	1.7
^{84}Se	8.6797	0.4048	6.2×10^{9}	7.0	3.6
^{82}Ge	8.5964	0.3902	1.7×10^{10}	9.8	3.9
^{80}Zn	8.4675	0.3750	3.8×10^{10}	12.5	4.3
^{78}Ni	8.2873	0.3590	1.3×10^{11}	18.5	4.7
^{76}Fe	7.9967	0.3421	2.2×10^{11}	21.5	5.2
^{74}Cr	7.6361	0.3243	3.1×10^{11}	24.0	5.9
^{118}Kr	7.2002	0.3051	(3.7×10^{11})	(25.0)	...

NOTE.—b is the binding energy per nucleon; $\rho_{\rm max}$ is the maximum density at which the nuclide is present; μ_e is the electron chemical potential at that density, and $\Delta\rho/\rho$ is the fractional increase in the mass density in the transition to the next nuclide. The value of $\rho_{\rm max} = 3.7 \times 10^{11}$ g cm^{-3} is the density at which neutron drip begins.

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TABLE 4

THE EQUILIBRIUM NUCLEI CALCULATED BY SALPETER (1961)

Nucleus	ρ_{max} (g cm^{-3})	μ_e (MeV)	Nucleus	ρ_{max} (g cm^{-3})	μ_e (MeV)
^{56}Fe.....	1.4×10^{7}	1.1	^{76}Zn.....	1.3×10^{10}	9.0
^{62}Ni.....	4.3×10^{8}	3.0	^{78}Zn.....	1.9×10^{10}	10.0
^{64}Ni.....	1.4×10^{9}	4.4	^{80}Zn.....	6.9×10^{10}	15.3
^{66}Ni.....	4.9×10^{9}	6.6	^{90}Ge.....	1.9×10^{11}	21.1
^{68}Ni.....	7.4×10^{9}	7.5	^{120}Sr.....	(3.4×10^{11})	(24.5)

at fixed n_b implies that, for $\Delta A \ll A$ and $\Delta Z \ll Z$,

$$\Delta\left(\frac{W_N + W_L}{A}\right) + \left(\frac{\Delta Z}{A} - \frac{Z}{A^2}\Delta A\right)\mu_e \geq 0 . \tag{21}$$

At fixed n_b the lattice energy W_L scales as $Z^2/A^{1/3}$ so that

$$\Delta W_L = W_L\left(\frac{2\Delta Z}{Z} - \frac{\Delta A}{3A}\right). \tag{22}$$

Substituting equations (22) and (18) into equation (21), we find

$$\Delta W_N \geq \mu\Delta A - \left(\mu_e + \frac{2W_L}{Z}\right)\Delta Z . \tag{23}$$

Consider first $\Delta Z = 0$, corresponding to changing the number of neutrons in nuclei; inequality (23) then becomes

$$\Delta_A W_N \geq \mu\Delta A . \tag{24}$$

This says that the chemical potential μ is always less than or equal to the energy (per neutron) required to add one or several neutrons to the nuclei. In the approximation that A may be treated as a continuous variable, inequality (24) reduces to the statement $\mu = \mu_n$, where $\mu_n = \partial W_N/\partial A$ is the neutron chemical potential (including the rest mass), or minus the neutron removal energy. If we keep A fixed and allow Z to change by ΔZ (multiple β-decays), inequality (23) becomes

$$\Delta_Z W_N \geq -(\mu_e + 2W_L/Z)\Delta Z . \tag{25}$$

This is the condition for beta stability of the nuclei; it is the analog of the usual β-stability condition

$$\Delta_Z W_N \geq -m_e c^2 \Delta Z \tag{26}$$

for isolated nuclei.

Neutron drip begins when the chemical potential $\mu - m_n c^2$ reaches $\epsilon_{n,0}$, the energy of the lowest neutron continuum state in the lattice. Consider, before free neutrons appear, increasing the baryon number of the entire system by A; this can be done either by adding a nucleus (A, Z) plus Z electrons to the system, which costs energy

$$\left(\frac{\partial E_{tot}}{\partial n_N}\right)_{n_e = Z n_N} = \mu A , \tag{27}$$

or by adding A neutrons in continuum states, which increases the energy by $(m_n c^2 + \epsilon_{n,0})A$. Before drip, $\mu < \epsilon_{n,0} + m_n c^2$. When these two energies first become equal, the free-neutron states will begin to be populated. The energy $\epsilon_{n,0}$ differs slightly from zero

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because of the interaction of a continuum neutron with the nuclei; it may be estimated by noting that one expects the average potential of a slow neutron passing through a nucleus such as ^{118}Kr to be attractive and on the order of a few tens of MeV; since at neutron drip the nuclei occupy about 10^{-3} [$=\rho/$(nuclear-matter mass density) $= \rho/(3 \times 10^{14}$ g cm^{-3})] of the volume of space, the average neutron potential is $\sim -10^{-2}$ MeV $\approx \epsilon_{n,0}$. If one neglects $\epsilon_{n,0}$, as we do here, the condition for neutron drip is that $\mu - m_n c^2 = 0$. We find that neutron drip begins at $\rho = 4.3 \times 10^{11}$ g cm^{-3}, at which point the electron chemical potential is 26.2 MeV, and the equilibrium nucleus is ^{118}Kr; this remains the most favorable nucleus somewhat beyond this point.

In Table 5 we list the equation of state of the ground state of matter. In this table ρ is the mass density, equal to the total energy density $E_{\rm tot}$ divided by c^2; n_b is the density of baryons, and the pressure is given by $P = n_b^2 \partial(E_{\rm tot}/n_b)/\partial n_b$. The general-relativistic adiabatic index Γ is defined by

$$\Gamma \equiv \frac{n_b}{P}\frac{\partial P}{\partial n_b} = \frac{\rho + P/c^2}{P}\frac{\partial P}{\partial \rho} . \tag{28}$$

The equation of state below $\rho = 10^4$ g cm^{-3} is that of Feynman *et al.* as tabulated in Harrison *et al.* (1965) (hereafter referred to as HTWW). The equation of state between 10^4 and 4.3×10^{11} g cm^{-3} is that calculated here; between 4.3×10^{11} and 5×10^{14} g cm^{-3}, it is that due to BBP. The extension of the equation of state to densities beyond 5×10^{14} g cm^{-3} is based on Pandharipande's model C calculation (1971), which allows for the presence of hyperons. We also list a second equation of state in this regime, also due to Pandharipande (1971), which is calculated for a pure neutron gas.

III. NONROTATING STARS

We proceed now to construct models for nonrotating stars composed of cold, fully catalyzed matter. This is done by integrating the general-relativistic equation of hydrostatic balance, the Tolman-Oppenheimer-Volkoff (TOV) equation (see HTWW 1965):

$$\frac{dP(r)}{dr} = -\frac{G[\rho(r) + P(r)/c^2][m(r) + 4\pi r^3 P(r)/c^2]}{r^2[1 - 2Gm(r)/rc^2]} , \tag{29}$$

$$m(r) = \int_0^r \rho(r') d^3r' \tag{30}$$

is the mass of the star within radial coordinate r and $P(r)$ is the pressure. Similarly, the number of baryons within radius r is given by

$$b(r) = \int_0^r \frac{n_b(r') d^3r'}{(1 - 2Gm(r')/r'c^2)^{1/2}} . \tag{31}$$

The radius R of the star is defined by

$$P(R) = 0 ; \tag{32}$$

the total mass M of the star (i.e., the gravitational mass observed by a distant spectator) is given by

$$M = m(R) = \int d^3r \rho(r) , \tag{33}$$

and the total number of baryons B in the star is given by

$$B = b(R) = \int \frac{d^3r n_b(r)}{[1 - 2Gm(r)/rc^2]^{1/2}} . \tag{34}$$

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TABLE 5

EQUATION OF STATE OF MATTER IN ITS GROUND STATE

ρ (g cm^{-3})	P (dynes cm^{-2})	n_b (cm^{-3})	Z	A	Γ
		Feynman-Metropolis-Teller			
7.86.....	$\leq$1.01*E*9	4.73*E*24	26	56	...
7.90.....	1.01*E*10	4.76*E*24	26	56	...
8.15.....	1.01*E*11	4.91*E*24	26	56	...
11.6.....	1.21*E*12	6.99*E*24	26	56	...
16.4.....	1.40*E*13	9.90*E*24	26	56	...
45.1.....	1.70*E*14	2.72*E*25	26	56	...
212.....	5.82*E*15	1.27*E*26	26	56	...
1150.....	1.90*E*17	6.93*E*26	26	56	...
		Present Calculation			
1.044*E*4.....	9.744*E*18	6.295*E*27	26	56	1.796
2.622*E*4.....	4.968*E*19	1.581*E*28	26	56	1.744
6.587*E*4.....	2.431*E*20	3.972*E*28	26	56	1.706
1.654*E*5.....	1.151*E*21	9.976*E*28	26	56	1.670
4.156*E*5.....	5.266*E*21	2.506*E*29	26	56	1.631
1.044*E*6.....	2.318*E*22	6.294*E*29	26	56	1.586
2.622*E*6.....	9.755*E*22	1.581*E*30	26	56	1.534
6.588*E*6.....	3.911*E*23	3.972*E*30	26	56	1.482
8.293*E*6.....	5.259*E*23	5.000*E*30	28	62	1.471
1.655*E*7.....	1.435*E*24	9.976*E*30	28	62	1.437
3.302*E*7.....	3.833*E*24	1.990*E*31	28	62	1.408
6.589*E*7.....	1.006*E*25	3.972*E*31	28	62	1.386
1.315*E*8.....	2.604*E*25	7.924*E*31	28	62	1.369
2.624*E*8.....	6.676*E*25	1.581*E*32	28	62	1.357
3.304*E*8.....	8.738*E*25	1.990*E*32	28	64	1.355
5.237*E*8.....	1.629*E*26	3.155*E*32	28	64	1.350
8.301*E*8.....	3.029*E*26	5.000*E*32	28	64	1.346
1.045*E*9.....	4.129*E*26	6.294*E*32	28	64	1.344
1.316*E*9.....	5.036*E*26	7.924*E*32	34	84	1.343
1.657*E*9.....	6.860*E*26	9.976*E*32	34	84	1.342
2.626*E*9.....	1.272*E*27	1.581*E*33	34	84	1.340
4.164*E*9.....	2.356*E*27	2.506*E*33	34	84	1.338
6.601*E*9.....	4.362*E*27	3.972*E*33	34	84	1.337
8.312*E*9.....	5.662*E*27	5.000*E*33	32	82	1.336
1.046*E*10.....	7.702*E*27	6.294*E*33	32	82	1.336
1.318*E*10.....	1.048*E*28	7.924*E*33	32	82	1.336
1.659*E*10.....	1.425*E*28	9.976*E*33	32	82	1.335
2.090*E*10.....	1.938*E*28	1.256*E*34	32	82	1.335
2.631*E*10.....	2.503*E*28	1.581*E*34	30	80	1.335
3.313*E*10.....	3.404*E*28	1.990*E*34	30	80	1.335
4.172*E*10.....	4.628*E*28	2.506*E*34	30	80	1.334
5.254*E*10.....	5.949*E*28	3.155*E*34	28	78	1.334
6.617*E*10.....	8.089*E*28	3.972*E*34	28	78	1.334
8.332*E*10.....	1.100*E*29	5.000*E*34	28	78	1.334
1.049*E*11.....	1.495*E*29	6.294*E*34	28	78	1.334
1.322*E*11.....	2.033*E*29	7.924*E*34	28	78	1.334
1.664*E*11.....	2.597*E*29	9.976*E*34	26	76	1.334
1.844*E*11.....	2.892*E*29	1.105*E*35	42	124	1.334
2.096*E*11.....	3.290*E*29	1.256*E*35	40	122	1.334
2.640*E*11.....	4.473*E*29	1.581*E*35	40	122	1.334
3.325*E*11.....	5.816*E*29	1.990*E*35	38	120	1.334
4.188*E*11.....	7.538*E*29	2.506*E*35	36	118	1.334
4.299*E*11.....	7.805*E*29	2.572*E*35	36	118	1.334

NOTE.—ρ is the mass density; P the pressure; n_b the baryon density; A and Z are the nucleon number and proton number, respectively, of the equilibrium nucleus present; Γ is the adiabatic index.

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TABLE 5—*Continued*

ρ (g cm^{-3})	P (dynes cm^{-2})	n_b (cm^{-3})	Z	A	Γ
Baym-Bethe-Pethick Equation of State					
4.460*E*11	7.890*E*29	2.670*E*35	40	126	0.40
5.228*E*11	8.352*E*29	3.126*E*35	40	128	0.36
6.610*E*11	9.098*E*29	3.951*E*35	40	130	0.40
7.964*E*11	9.831*E*29	4.759*E*35	41	132	0.46
9.728*E*11	1.083*E*30	5.812*E*35	41	135	0.54
1.196*E*12	1.218*E*30	7.143*E*35	42	137	0.63
1.471*E*12	1.399*E*30	8.786*E*35	42	140	0.73
1.805*E*12	1.638*E*30	1.077*E*36	43	142	0.83
2.202*E*12	1.950*E*30	1.314*E*36	43	146	0.93
2.930*E*12	2.592*E*30	1.748*E*36	44	151	1.06
3.833*E*12	3.506*E*30	2.287*E*36	45	156	1.17
4.933*E*12	4.771*E*30	2.942*E*36	46	163	1.25
6.248*E*12	6.481*E*30	3.726*E*36	48	170	1.31
7.801*E*12	8.748*E*30	4.650*E*36	49	178	1.36
9.611*E*12	1.170*E*31	5.728*E*36	50	186	1.39
1.246*E*13	1.695*E*31	7.424*E*36	52	200	1.43
1.496*E*13	2.209*E*31	8.907*E*36	54	211	1.44
1.778*E*13	2.848*E*31	1.059*E*37	56	223	1.46
2.210*E*13	3.931*E*31	1.315*E*37	58	241	1.47
2.988*E*13	6.178*E*31	1.777*E*37	63	275	1.49
3.767*E*13	8.774*E*31	2.239*E*37	67	311	1.51
5.081*E*13	1.386*E*32	3.017*E*37	74	375	1.53
6.193*E*13	1.882*E*32	3.675*E*37	79	435	1.54
7.732*E*13	2.662*E*32	4.585*E*37	88	529	1.56
9.826*E*13	3.897*E*32	5.821*E*37	100	683	1.60
1.262*E*14	5.861*E*32	7.468*E*37	117	947	1.65
1.586*E*14	8.595*E*32	9.371*E*37	143	1390	1.70
2.004*E*14	1.286*E*33	1.182*E*38	201	2500	1.74
2.520*E*14	1.900*E*33	1.484*E*38	...	...	1.81
2.761*E*14	2.242*E*33	1.625*E*38	...	...	1.82
3.085*E*14	2.751*E*33	1.814*E*38	...	...	1.87
3.433*E*14	3.369*E*33	2.017*E*38	...	...	1.92
3.885*E*14	4.286*E*33	2.280*E*38	...	...	1.97
4.636*E*14	6.103*E*33	2.715*E*38	...	...	2.03
5.094*E*14	7.391*E*33	2.979*E*38	...	...	2.05
Pandharipande Equation of State C for Hyperonic Matter					
6.968*E*14	2.720*E*34	4.000*E*38	...	...	...
8.797*E*14	3.789*E*34	5.000*E*38	...	...	...
1.066*E*15	5.478*E*34	6.000*E*38	...	...	...
1.448*E*15	1.110*E*35	8.000*E*38	...	...	...
1.854*E*15	2.153*E*35	1.000*E*39	...	...	...
2.399*E*15	4.108*E*35	1.250*E*39	...	...	...
3.125*E*15	7.522*E*35	1.550*E*39	...	...	...
4.086*E*15	1.360*E*36	1.900*E*39	...	...	...
5.369*E*15	2.379*E*36	2.300*E*39	...	...	...
7.740*E*15	4.616*E*36	2.900*E*39	...	...	...
1.129*E*16	8.437*E*36	3.600*E*39	...	...	...
1.729*E*16	1.525*E*37	4.500*E*39	...	...	...
2.607*E*16	2.574*E*37	5.500*E*39	...	...	...
3.728*E*16	3.943*E*37	6.500*E*39	...	...	...
5.112*E*16	5.645*E*37	7.500*E*39	...	...	...
Pandharipande Equation of State for Pure-Neutron Matter					
7.004*E*14	3.235*E*34	4.000*E*38	...	...	...
8.877*E*14	5.958*E*34	5.000*E*38	...	...	...
1.082*E*15	9.903*E*34	6.000*E*38	...	...	...
1.498*E*15	2.188*E*35	8.000*E*38	...	...	...
1.955*E*15	4.029*E*35	1.000*E*39	...	...	...
2.597*E*15	7.355*E*35	1.250*E*39	...	...	...
3.483*E*15	1.295*E*36	1.550*E*39	...	...	...
4.695*E*15	2.195*E*36	1.900*E*39	...	...	...
6.342*E*15	3.585*E*36	2.300*E*39	...	...	...
9.392*E*15	6.389*E*36	2.900*E*39	...	...	...
1.391*E*16	1.086*E*37	3.600*E*39	...	...	...
2.135*E*16	1.862*E*37	4.500*E*39	...	...	...
3.197*E*16	3.004*E*37	5.500*E*39	...	...	...
4.518*E*16	4.447*E*37	6.500*E*39	...	...	...
6.115*E*16	6.208*E*37	7.500*E*39	...	...	...

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The mass M contains both the internal energy and the gravitational self-energy of the star; this is the reason for the absence of the factor $(1 - 2Gm/rc^2)^{1/2}$ in the denominator of (33).

Equations (29) and (30) are readily integrated numerically, for a given equation of state, to construct the pressure profile $P(r)$, and from this the mass density profile $\rho(r)$ and baryon density profile $n_b(r)$. The single integration constant P_c, the pressure at the center of the star, completely specifies the solution. The actual numerical integration was done by a four-point Runge-Kutta sequence, in which the radial step size Δ was taken to be a fixed fraction δ of a mean mass-pressure scale height

$$\Delta = \delta\left(\frac{1}{m}\frac{dm}{dr} - \frac{1}{P}\frac{dP}{dr}\right)^{-1}. \tag{35}$$

The advantage of this prescription for Δ is that the step size is small when either m (for $r \ll R$) or P (for $r \lesssim R$) changes rapidly; furthermore, one does not need to have an estimate of the radius beforehand, as when using a fixed Δ. The value of $\delta = 0.23$ (or ten steps per pressure-mass decade) was found to be optimal from the points of view of both computing time and accuracy. An increase in the number of integration steps to 20 per pressure-mass decade ($\delta = 0.115$) produced only one part in 10^5 changes in computed masses and radii, while a decrease to 5 steps per decade ($\delta = 0.46$) produced changes of less than 1 percent. The value $\delta = 0.23$ gives $\sim$150–200 integration steps per star. Each integration step yields a value of $P(r)$, and the values of $\rho(r)$ and $n_b(r)$ were constructed from the numerical equation of state by linearly interpolating log ρ and log n_b as functions of log P.

Figure 1 shows the resulting stellar masses as a function of the central density $\rho_c = \rho(r = 0)$ in the range 10^7 to 5×10^{14} g cm^{-3}, the limit of reliability of our equation of state. The minimum-mass stable neutron star, corresponding to the minimum in M versus ρ_c, has a mass 0.0925 $M_\odot$ and a central density 1.55×10^{14} g cm^{-3}. Since in the calculations of BBP the crust of a neutron star dissolves at $\rho \sim 2 \times 10^{14}$ g cm^{-3}, neutron stars of mass in the range $0.0925 < M/M_\odot \lesssim 0.11$ are, for this equation of state, entirely solid! The maximum-mass neutron star for which we have a reliable equation of state is 0.17 $M_\odot$.

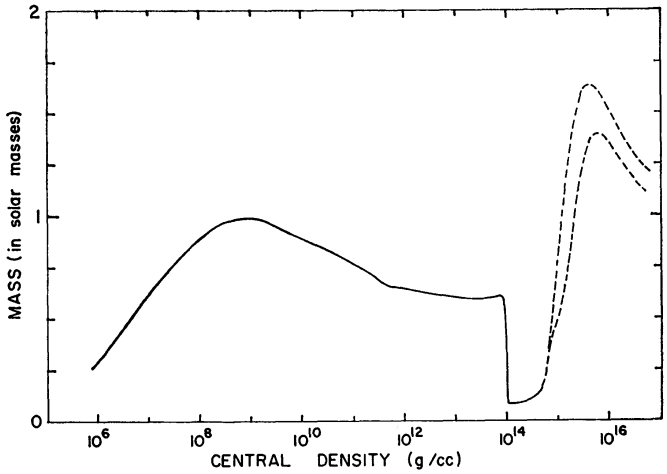


FIG. 1.—Mass versus central density for zero-temperature nonrotating stars in nuclear equilibrium. Stars to the left of the maximum (the Chandrasekhar limit) at $\rho_c = 1.4 \times 10^9$ g cm^{-3} are stable white dwarfs, while stars to the right of the minimum at 1.55×10^{14} g cm^{-3} are neutron stars. The lower dashed extension of the curve was constructed from Pandharipande's hyperonic equation of state C, and the upper dashed extension from Pandharipande's equation of state for pure neutrons. Neutron stars beyond the maximum are unstable.

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The mass and central density of the minimum-mass neutron star are not so very different from the values of 1.26×10^{14} g cm^{-3} and 0.065 $M_\odot$ obtained by Cohen and Cameron (1970), who used an equation of state calculated from the Levinger-Simmons V_γ potential (Langer *et al.* 1969). At higher densities the mass of our models increases rather slowly, reaching 0.17 $M_\odot$ for a central density of 5×10^{14} g cm^{-3}; at this central density the mass of a V_γ model is 1.13 $M_\odot$. This difference is due to the fact that for mass densities greater than about 10^{14} g cm^{-3} the V_γ equation of state is very much harder than ours. (At a density of 5×10^{14} g cm^{-3} the V_γ equation of state gives a pressure about 6 times larger than the BBP equation of state.) Our calculated masses agree with those obtained by Wang, Rose, and Schlenker (1970) in the range of central densities above 3×10^{14} g cm^{-3} where they find stable stars. This is not surprising, since Wang *et al.*'s equation of state is similar to that of BBP in this region. The lower dashed section of the curve in Figure 1, from 5×10^{14} to 10^{16} g cm^{-3}, was constructed by using Pandharipande's hyperonic extension C of the equation of state to very high densities. With this extension of the equation of state one finds a maximum mass of 1.41 $M_\odot$ for a stable neutron star, with a central density 6.2×10^{15} g cm^{-3}.[1] The upper dashed curve results from using Pandharipande's pure-neutron equation of state above 5×10^{14} g cm^{-3}; the result is to increase the maximum mass to 1.66 $M_\odot$ for a stable neutron star, but to reduce its central density to 4.1×10^{15} g cm^{-3}.

The maximum white-dwarf mass, the Chandrasekhar limit, is for this equation of state 1.00 $M_\odot$, quite close to the value 1.015 $M_\odot$ given by Hamada and Salpeter (1961). The central density of this star is 1.4×10^9 g cm^{-3}; it consists of ^{56}Fe, ^{62}Ni, and ^{64}Ni with a central core of ^{84}Se (all laboratory nuclei). Stars of central densities between 1.4×10^9 and 1.55×10^{14} g cm^{-3} are unstable against radial oscillations (HTWW 1965). While one does not expect white dwarfs with less than a solar mass ever to reach complete nuclear equilibrium, these models give one an indication of the character of the cores of heavier white dwarfs.

A necessary condition for stability nonrelativistically is that an average of $\Gamma - \frac{4}{3}$ over the star be positive. Relativistically this condition is modified to the requirement than an average of $\Gamma - \frac{4}{3}$ exceed KR_S/R, where $R_S = 2MG/c^2$ is the Schwarzschild radius of the star, R is the radius of the star, and K is a dimensionless number of order unity (see Chandrasekhar 1964). Before neutron drip, $\Gamma - \frac{4}{3}$ is always positive. Nonrelativistically, pure ^{56}Fe white dwarfs are stable at any central density. Calculated relativistically, they are stable up to a central density 3×10^{10} g cm^{-3}, for which $M = 1.15\ M_\odot$, $R = 910$ km, and $R_S = 3.39$ km. At this central density $\Gamma - \frac{4}{3} = 0.0012$, which is comparable to $R_S/R = 0.0037$; the instability of stars with greater central density is a general-relativistic effect. On the other hand, the instability above 1.4×10^9 g cm^{-3} of the white dwarfs calculated here is a consequence of the phase transition from ^{64}Ni to ^{84}Se which occurs at $\rho = 1.2 \times 10^9$ g cm^{-3}, with a relatively large density discontinuity of 7.9 percent. The point is that the average of Γ near the phase transition is effectively lowered, since across the transition a discontinuous increase in the baryon number produces no increase in the pressure; phase transitions tend to make the matter softer. The effective reduction in the average of Γ is $\sim\Delta\rho/\rho$; by the ^{64}Ni–^{84}Se transition Γ is sufficiently close to $\frac{4}{3}$ that the large $\Delta\rho/\rho$ there makes stars with central densities only slightly above the transition density unstable.

[1] We have also calculated neutron-star models by using Pandharipande's model A equation of state, for densities greater than 5×10^{14} g cm^{-3}. In this model no reduction in the intermediate-range attraction is made; this attraction leads to a phase transition in the matter, in which the density jumps discontinuously from $\sim 2.6 \times 10^{14}$ to $\sim 1.4 \times 10^{15}$ g cm^{-3}. The resulting stellar models with central densities between 1.4×10^{15} and 1.8×10^{15} g cm^{-3} are unstable. For this equation of state the maximum mass of a stable neutron star is 1.27 $M_\odot$; the corresponding central density is 8.1×10^{15} g cm^{-3}. Comparing the stellar models for Pandharipande's three extensions of the equation of state (A, C, and pure neutron), we observe the general rule that the stiffer the equation of state, the higher the mass of the most massive stable neutron star and the smaller its central density.

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Figure 2 shows the mass versus radius for the present models. The maximum radius for a stable neutron star is ~164 km, while the white dwarf with $M = 1.00\ M_\odot$ has a radius 2140 km. Table 6 gives the radius, mass, and total baryon number of these models for a selected set of central densities. Included in Table 6 is z_s, the surface redshift, given by

$$z_s = \frac{1}{(1 - 2GM/Rc^2)^{1/2}} - 1\ ; \tag{36}$$

also given is b', the binding energy per baryon, measured with respect to m_H, the hydrogen atom mass:

$$b' \equiv -\left(\frac{Mc^2}{B} - m_H c^2\right). \tag{37}$$

Measured this way, the binding energy per nucleon of an atom of ^{56}Fe is 8.37 MeV.

In Figure 3 we present the binding energy as a function of B, the total baryon number for these models. Included are the models for $B \geq 2.5 \times 10^{56}$, which are constructed by using Pandharipande's hyperonic extension C of the equation of state, as well as, for comparison, the curve for $B \lesssim 10^2$, the range of ordinary nuclei. Figure 3 is basically a graph of the ground-state energy of a collection of B baryons, including all interactions—strong, electromagnetic, weak, and gravitational. Up to the Chandrasekhar limit for white dwarfs the gravitational binding energy is insignificant compared with nuclear binding; the slight upswing of b' in the white dwarf regime is due to gravity. Up to $B = 1.9 \times 10^{56}$, a neutron star of a given B is less strongly bound than a white dwarf of

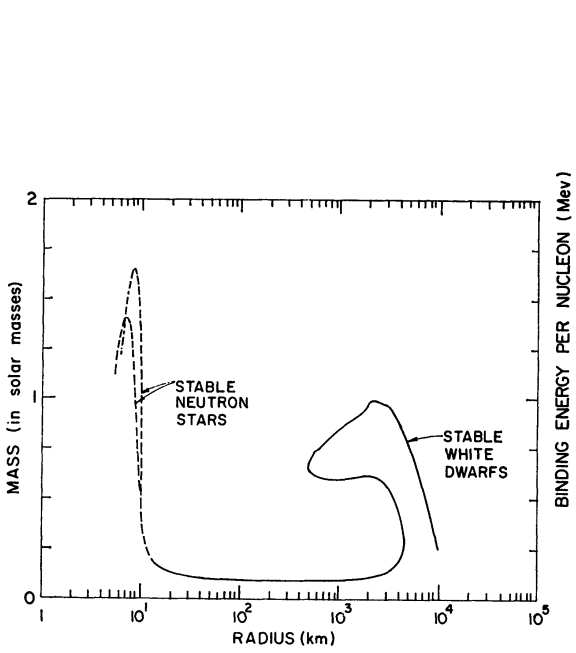


FIG 2

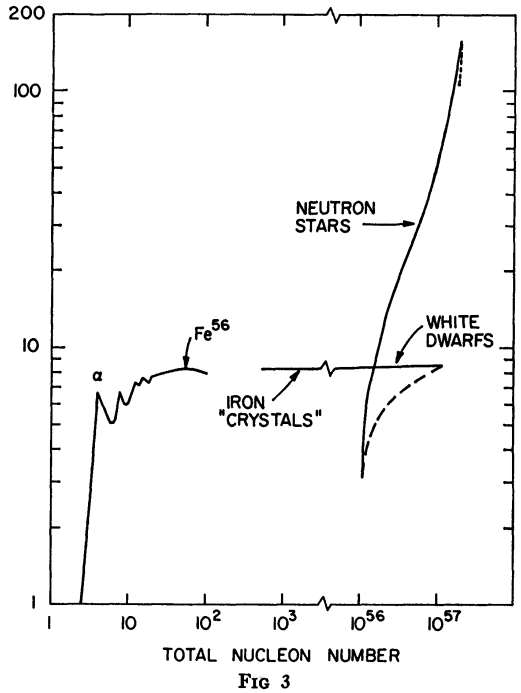


FIG 3

FIG. 2.—Mass versus radius for zero-temperature stars in nuclear equilibrium. Stable portions of the curve terminate at the extrema. Upper and lower dashed curves are as in Fig. 1.

FIG. 3.—Binding energy per nucleon of a collection of baryons in the ground state, including strong, electromagnetic, weak, and gravitational interactions. The binding energy is defined as minus the total energy per baryon, measured with respect to the hydrogen atom mass. Ordinary nuclei are on the left, white dwarfs and neutron stars on the right. Dotted portions of the curve represent unstable stars. Above 1.9×10^{56} baryons, neutron stars are more strongly bound than white dwarfs.

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TABLE 6

NONROTATING WHITE-DWARF AND NEUTRON-STAR MODELS

ρ_c (g cm^{-2})	R (km)	$M/M_\odot$	B	z_s	b' (MeV)
4.33×10^{6}	7380	0.502	6.02×10^{56}	1.00×10^{-4}	8.41
2.97×10^{7}	5110	0.759	9.10×10^{56}	2.19×10^{-4}	8.45
1.38×10^{9}	2140	1.00	1.20×10^{57}	6.92×10^{-4}	8.50
3.01×10^{10}*	944	0.845	1.01×10^{57}	1.32×10^{-3}	8.37
1.00×10^{14}*	2620	0.612	7.34×10^{56}	3.45×10^{-4}	7.71
1.55×10^{14}	164	0.0925	1.10×10^{56}	8.34×10^{-4}	3.17
2.00×10^{14}	56.5	0.100	1.19×10^{56}	2.62×10^{-3}	3.62
2.30×10^{14}	40.9	0.105	1.26×10^{56}	3.81×10^{-3}	4.02
2.60×10^{14}	32.9	0.111	1.33×10^{56}	5.02×10^{-3}	4.43
2.90×10^{14}	27.7	0.117	1.40×10^{56}	6.30×10^{-3}	4.89
3.20×10^{14}	24.2	0.124	1.48×10^{56}	7.65×10^{-3}	5.39
3.60×10^{14}	20.9	0.132	1.59×10^{56}	9.46×10^{-3}	6.12
4.00×10^{14}	18.6	0.142	1.70×10^{56}	1.15×10^{-2}	6.93
4.50×10^{14}	16.5	0.154	1.85×10^{56}	1.41×10^{-2}	8.02
5.00×10^{14}	15.0	0.167	2.00×10^{56}	1.69×10^{-2}	9.18
6.00×10^{14}	11.5	0.244	2.95×10^{56}	3.29×10^{-2}	15.7
8.00×10^{14}	10.1	0.424	5.20×10^{56}	6.84×10^{-2}	29.4
1.00×10^{15}	9.8	0.516	6.38×10^{56}	8.82×10^{-2}	36.4
3.00×10^{15}	8.1	1.25	1.68×10^{57}	3.54×10^{-1}	109
6.20×10^{15}	7.0	1.41	1.96×10^{57}	5.74×10^{-1}	134
1.00×10^{16}*	6.4	1.37	1.88×10^{57}	6.49×10^{-1}	123

NOTE.—ρ_c is the central density, R is the radius, $M/M_\odot$ is the mass in solar masses, B is the total number of baryons in the star, z_s is the surface redshift ($\delta\lambda/\lambda$), and b' is the binding energy per baryon, measured with respect to the hydrogen atom mass. Stars indicated with an asterisk are unstable against radial oscillations. For $\rho_c > 5 \times 10^{14}$, Pandharipande's equation of state C was used.

the same B. The decrease in the binding energy in light neutron stars, to $\sim$3 MeV, is due microscopically to the fact that the matter is neutron-rich; neutron-neutron interactions are less attractive than neutron-proton interactions. In addition, the exclusion principle raises the mean kinetic energy per nucleon inside the neutron-rich nuclei. As one goes to higher-mass neutron stars, the gravitational binding energy per nucleon increases rapidly. The rapid rise in binding energy shown here for the heavier neutron stars is qualitatively correct, but the details are uncertain, due to the lack of knowledge of the equation of state at high densities. Note that the binding energy and mass are always extremal at the same points (HTWW 1965, theorem 16, p. 61).

In Figure 4 we show density profiles of several representative neutron stars. An interesting feature of these profiles, compared with those generated (Cohen *et al.* 1970) from the V_γ equation of state (Langer *et al.* 1969; Langer and Rosen 1970) is that the rise in density above neutron drip is more gradual here. This is because the present equation of state is harder (i.e., the matter is less compressible) at low densities (10^{12} g cm$^{-3} < \rho < 10^{13}$ g cm^{-3}) and softer at higher densities than the V_γ equation of state.

We conclude this section by describing several checks that were made on the accuracy of the integration routine. The first was a comparison with the analytic results for a star composed of incompressible matter. For a star of constant density ρ and central pressure P_c, the TOV equation can be integrated exactly to give a pressure profile

$$P(r) = \rho c^2\left[\frac{2}{3 - (1 - 8\pi r^2\rho G/3c^2)^{1/2}} - 1\right] \tag{38}$$

The radius R is given by

$$R = \left(\frac{3P_c}{2\pi\rho G}\right)^{1/2} \frac{(1 + 2P_c/\rho c^2)^{1/2}}{1 + 3P_c/\rho c^2}, \tag{39}$$

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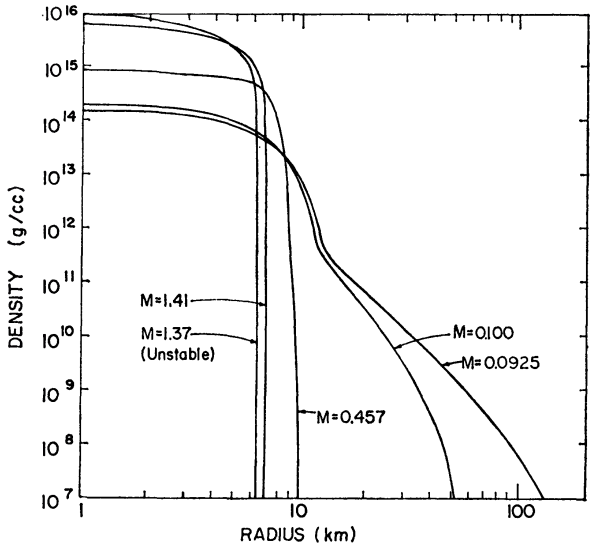


FIG. 4.—Density profiles for five neutron stars. Masses are measured in solar masses. The star with $M = 0.0925\ M_\odot$ is the lightest stable neutron star; that with $M = 1.41\ M_\odot$ is the heaviest stable neutron star found by using Pandharipande's hyperonic extension C of the equation of state. The two lightest stars are entirely solid.

and the mass by $M = \frac{4}{3}\pi\rho R^3$. For $P_c/\rho c^2 \to \infty$, $R \to (c^2/3\pi G)^{1/2}$. As a test we numerically integrated the TOV equation for several values of P_c, both with Δ fixed (at about $R/150$) and with Δ given by equation (35). The agreement between computed values of R and the analytic expression was better than one part in 10^5.

As a second check, the alternative expression for the stellar mass (HTWW 1965)

$$M = \int_0^R d^3r e^{\nu(r)/2} \frac{[\rho(r) + 3P(r)/c^2]}{[1 - 2Gm(r)/rc^2]^{1/2}} \tag{40}$$

was also calculated. Here $e^{\nu(r)} = -g_{00}$ is given in terms of the distribution of matter by the equilibrium condition

$$e^{\nu(r)/2}\mu(r) = \text{constant} = \mu(R)(1 - 2GM/Rc^2)^{1/2}, \tag{41}$$

where

$$\mu(r) = \frac{\rho(r)c^2 + P(r)}{n_b(r)} \tag{42}$$

is the local neutron chemical potential (including the rest mass). The results of integrating equation (40) agreed with those obtained from equation (33) to better than one part in 10^5.

IV. SLOWLY ROTATING STARS

The moment of inertia of a slowly rotating star is readily calculated[2] in terms of the mass distribution of the nonrotating star. To do this, one assumes the star to be rotating slowly with a uniform angular velocity Ω, as seen by a distant observer. The angular velocity $\bar{\omega}(r)$ of a point in the star measured (by this observer) with respect to the angular velocity of the local inertial frame is determined by the equation

$$\frac{1}{r^4}\frac{d}{dr}\left(r^4 j \frac{d\bar{\omega}}{dr}\right) + \frac{4}{r}\frac{dj}{dr}\bar{\omega} = 0, \tag{43}$$

[2] The general-relativistic theory of slowly rotating stars is reviewed by Thorne (1969) and Cohen (1970).

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where

$$j = e^{-\nu(r)/2}(1 - 2Gm(r)/rc^2)^{1/2} ; \tag{44}$$

the linear equation (43) is integrated (numerically) subject to the boundary conditions that $\tilde{\omega}(r)$ is regular as $r \to 0$, and that $\tilde{\omega}(r) \to \Omega$ as $r \to \infty$. The total angular momentum of the star is then given by

$$J = \frac{c^2}{6G} R^4 \left(\frac{d\tilde{\omega}}{dr}\right)_{r=R} , \tag{45}$$

where R is the radius of the (nonrotating) star. The moment of inertia of the star is given by

$$I = J/\Omega , \tag{46}$$

as in the nonrelativistic case.

The moment of inertia can be equivalently calculated from the expression

$$I = \frac{8\pi}{3} \int_0^R dr \frac{r^4(\rho(r) + P(r)/c^2)}{[1 - 2Gm(r)/rc^2]^{1/2}} \frac{\tilde{\omega}(r)}{\Omega} e^{-\nu(r)/2} . \tag{47}$$

The moments of inertia for several neutron stars are listed in Table 7, as functions of their central densities. The two methods of calculation, (46) and (47), yield results identical to one part in 10^5. It is interesting to note that as a consequence of general relativity the moment of inertia of the more massive neutron stars can considerably exceed $\frac{2}{5}MR^2$. Also given in Table 7 are the rates of dragging of the local inertial frames (as seen from infinity)

$$\omega(r) = \Omega - \tilde{\omega} \tag{48}$$

for the center (ω_c) and surface (ω_s) of the star.

TABLE 7

ROTATING NEUTRON-STAR MODELS

ρ_c (g cm^{-3})	Ω_c (s^{-1})	I (g-cm^2)	ω_s/Ω	ω_c/Ω	ϵ	q/MR^2	$\delta M/M$
1.00×10^{14}*	2.1	7.38×10^{48}	6.09×10^{-5}	2.92×10^{-2}	0.854	1.30×10^{-2}	1.21×10^{-1}
1.55×10^{14}	52.7	1.42×10^{44}	4.79×10^{-6}	3.46×10^{-2}	1.00	1.61×10^{-4}	2.15×10^{-3}
2.00×10^{14}	271	6.23×10^{43}	5.12×10^{-5}	4.05×10^{-2}	1.00	1.47×10^{-4}	4.14×10^{-3}
2.30×10^{14}	452	5.75×10^{43}	1.25×10^{-4}	4.43×10^{-2}	1.00	2.59×10^{-4}	7.51×10^{-3}
2.60×10^{14}	644	5.59×10^{43}	2.33×10^{-4}	4.80×10^{-2}	1.00	4.78×10^{-4}	1.20×10^{-2}
2.90×10^{14}	855	5.54×10^{43}	3.86×10^{-4}	5.18×10^{-2}	1.00	8.55×10^{-4}	1.80×10^{-2}
3.20×10^{14}	1.8×10^{3}	5.54×10^{43}	5.82×10^{-4}	5.57×10^{-2}	1.00	1.40×10^{-3}	2.47×10^{-2}
3.60×10^{14}	1.39×10^{3}	5.62×10^{43}	9.15×10^{-4}	6.10×10^{-2}	1.00	2.42×10^{-3}	3.47×10^{-2}
4.00×10^{14}	1.71×10^{3}	5.76×10^{43}	1.33×10^{-3}	6.65×10^{-2}	1.00	3.77×10^{-3}	4.51×10^{-2}
4.50×10^{14}	2.13×10^{3}	5.99×10^{43}	1.97×10^{-3}	7.35×10^{-2}	1.01	5.92×10^{-3}	5.94×10^{-2}
5.00×10^{14}	2.55×10^{3}	6.26×10^{43}	2.74×10^{-3}	8.06×10^{-2}	1.01	8.50×10^{-3}	7.40×10^{-2}
6.00×10^{14}	4.61×10^{3}	8.94×10^{43}	8.71×10^{-3}	1.17×10^{-1}	1.03	2.71×10^{-2}	1.50×10^{-1}
8.00×10^{14}	7.39×10^{3}	1.87×10^{44}	2.69×10^{-2}	1.85×10^{-1}	1.09	6.75×10^{-2}	2.65×10^{-1}
1.00×10^{15}	8.50×10^{3}	2.40×10^{44}	3.75×10^{-2}	2.24×10^{-1}	1.10	8.02×10^{-2}	2.94×10^{-1}
3.00×10^{15}	1.76×10^{4}	6.24×10^{44}	1.73×10^{-1}	5.97×10^{-1}	1.05	9.18×10^{-2}	2.60×10^{-1}
6.20×10^{15}	2.34×10^{4}	6.20×10^{44}	2.69×10^{-1}	8.01×10^{-1}	0.964	9.17×10^{-2}	1.89×10^{-1}
1.00×10^{16}*	2.64×10^{4}	5.20×10^{44}	2.96×10^{-1}	8.72×10^{-1}	0.933	9.15×10^{-2}	1.64×10^{-1}

NOTE.—ρ_c is the central density, $\Omega_c = (GM/R^3)^{1/2}$, I is the moment of inertia; ω_s/Ω is the fractional rate of dragging of the local inertial frame at the surface of the star (as seen from infinity), and ω_c/Ω is the corresponding rate for the center of the star, The parameter ϵ times $|\Omega|/\Omega_c$ is the surface eccentricity, and q times $(\Omega/\Omega_c)^2$ is the mass quadrupole moment. The quantity δM times $(\Omega/\Omega_c)^2$ gives the increase in mass of a rotating star, compared with a nonrotating star of the *same* central density. Stars indicated with an asterisk are unstable against radial oscillations. Static properties of stars of the same central density (and equation of state) are given in Table 6.

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The total mass of a slowly rotating star is related to the mass of the same star in the absence of rotation by (Hartle 1970)

$$M(B, \Omega)c^2 = M(B, O)c^2 + \tfrac{1}{2}I\Omega^2 , \tag{49}$$

where B is the number of baryons in the star; as in the nonrelativistic case, $\frac{1}{2}I\Omega^2$ is the rotational energy of the star.

The rotation produces both spherical and quadrupolar deformations of the star. In particular, the central density of the rotating star is changed; the central density of stable stars is lowered by rotation. For a fixed central density the mass of a rotating star is of the form

$$M(\rho_c ,\Omega) = M(\rho_c, 0) + \delta M(\Omega/\Omega_c)^2 , \tag{50}$$

where

$$\Omega_c \equiv [GM(\rho_c ,0)/R(\rho_c ,0)^3]^{1/2} ; \tag{51}$$

Ω_c is roughly the angular velocity at which the star begins to shed mass at the equator. The quantity δM may be calculated from equation (3-53) of Thorne (1969); the results are tabulated in Table 7.

The Newtonian gravitational potential very far from a rotating star has the form, to second order in Ω,

$$\Phi = -\frac{GM(B, \Omega)}{r} + \frac{P_2 (\cos \theta)}{r^3} q \left(\frac{\Omega}{\Omega_c}\right)^2 , \tag{52}$$

where

$$Q = q(\Omega/\Omega_c)^2 \tag{53}$$

is the quadrupole moment of the rotating star. The surface eccentricity of a rotating star, normally defined as

$$e = [1 - (\text{radius at pole})^2/(\text{radius at equator})^2]^{1/2} , \tag{54}$$

is proportional to $|\Omega|$ for small Ω; we write

$$e = \epsilon|\Omega|/\Omega_c . \tag{55}$$

To order Ω, the coefficients ϵ and q are the same for rotations at fixed central density as for rotations at fixed baryon number. Values of ϵ and q, calculated according to equations (3-59), (3-60), and (3-62) of Thorne (1969), are listed in Table 7, for stars of given central density.

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